# Homework 5

MATH 518 - Fall 2025

**Due on Friday, November 21, 11:59pm on Crowdmark**

Total of points = 30 points

Please submit each question separately on Crowdmark. Unless stated otherwise, $k$ is an algebraically closed field of characteristic zero.

1. (10 points) Let $X = V(x^3 - y^2) \subset k^2$. Is the map
$$f\colon k \longrightarrow X, \qquad f(t) = (t^2, t^3)$$
a finite map?

*The vector space of homogeneous polynomials in $k[x_1, \ldots, x_{n+1}]$ is of degree $d \geq 1$ is of dimension $N \coloneqq \binom{n+d}{n}$. It has a basis consisting of monomials $x^I \coloneqq x_1^{i_1} \cdots x_{n+1}^{i_{n+1}}$, where $I = (i_1, \ldots, i_{n+1})$ and $|I| = i_1 + \cdots + i_{n+1} = d$. For $d \geq 1$, we have a morphism*
$$\nu_d\colon \mathbb{P}^n(k) \longrightarrow \mathbb{P}^{N-1}(k)$$
$$p = [x_1 : \cdots : x_{n+1}] \longmapsto \nu_d(p) = [x_1^d : x_1^{d-1}x_2 : \cdots : x_{n+1}^d].$$
*The image consists of all possible monomials $x^I$ of degree $d$. It is an isomorphism onto its image, called the* Veronese embedding. *The goal of the next exercise is to show this in the special case $n = 1$ (to simplify the notation, the idea would be the same for $n > 1$).*

2. (10 points) When $n = 1$, the Veronese embedding is given by
$$\nu_d([x\colon y]) = [x^d : x^{d-1}y : \cdots : y^{d-1}x : y^d].$$
Show that $\nu_d$ is an isomorphism onto its image, and that the image is the projective variety $V = V_p(I)$ where $I \subset k[z_1, \ldots, z_{d+1}]$ is the homogeneous ideal $I \coloneqq \langle z_i z_j - z_k z_l \mid i + j = k + l \rangle$.

3. (10 points) For a homogeneous polynomial $f \in k[x_1, \ldots, x_{n+1}]$ of degree $d \geq 1$, let $U_f \subseteq \mathbb{P}^n(k)$ be the open subset
$$U_f \coloneqq \{p \in \mathbb{P}^n(k) \mid f(p) \neq 0\} = \mathbb{P}^n(k) \setminus V_p(f).$$

In the last lecture, we have seen that $U_f$ is an affine variety when $f = a_1x_1 + \cdots + a_{n+1}x_{n+1}$ is a linear polynomial. Use it to show that $U_f$ is affine for any $f$.
*Hint: use the Veronese embedding.*